Product Requirements Document (PRD)

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Personalized Study Assistant

1. Introduction

The Personalized Study Assistant is an AI-powered platform designed to enhance the learning experience by automatically generating study materials from user-uploaded documents. It leverages spaced repetition and adaptive testing to optimize knowledge retention.

2. Target Audience

- Students preparing for exams.
- Professionals acquiring new certifications.
- Lifelong learners seeking to understand complex material.

3. Core Features

3.1 User Management

- User registration and authentication.
- Profile management.

3.2 Document Management

- Uploading educational documents (PDFs, text, etc.).
- Automated text extraction and chunking (parsing large texts into manageable sections).

3.3 Automated Study Material Generation

- **Flashcards:** AI-generated flashcards based on document content, featuring a question (front) and answer (back).
- **Quizzes:** Automatically generated multiple-choice questions with answer explanations and source citations to ground the AI's reasoning.

3.4 Spaced Repetition System (SRS)

- A spaced repetition algorithm (e.g., SuperMemo-2 style) applied to flashcard reviews.
- Tracking of ease factor, interval, and repetitions to schedule the next review date dynamically.

3.5 Study Sessions and Analytics

- Structured study sessions testing the user on generated quizzes and due flashcards.
- Scoring and tracking of weak topics based on session performance.
- Adaptive learning paths that focus more on identified weak topics.

4. Data Model Overview

- **User:** Manages identity and relations to documents and study activities.
- **Document & DocumentChunk:** Stores uploaded content, split into smaller chunks for AI processing and citation mapping.
- **Flashcard:** Stores SRS metadata (nextReviewDate, easeFactor, repetitions) alongside card content.
- **QuizQuestion:** Stores multiple-choice structure, the correct answer, explanations, and citations linking back to the document.
- **StudySession:** Records the score and specifically flags weak topics to guide future sessions.

5. Technical Stack

- **Frontend/Framework:** Next.js (React)
- **Database ORM:** Prisma
- **Database Engine:** PostgreSQL
- **AI Integration:** LangChain, Google Generative AI (Gemini), OpenAI models (as needed).
- **Document Processing:** PDF parsing for text extraction.

6. Future Scope

- Collaboration features (sharing flashcards and documents).
- Gamification (streaks, badges, leaderboards).
- Voice-enabled Q&A sessions.